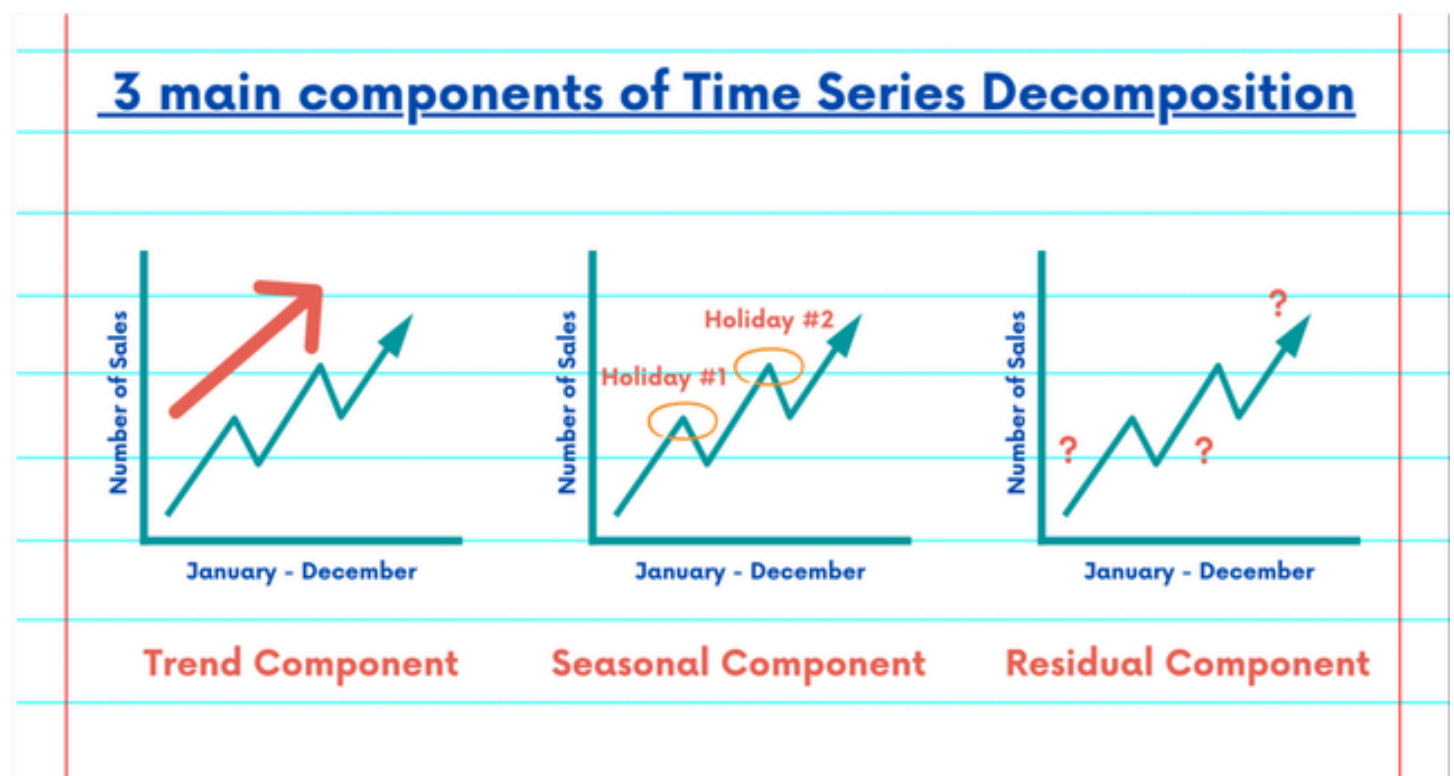


I. WHAT IS TIME SERIES DATA?

A Time Series is a sequence of data points indexed in successive order, usually with a uniform time interval (e.g., hourly stock prices, daily weather, or monthly sales). Unlike standard regression, Time Series data has Temporal Dependency—what happens today is heavily influenced by what happened yesterday.

1.1. The Three Components of Time Series

1. Trend: The long-term increase or decrease in the data (e.g., the global warming trend).
2. Seasonality: Patterns that repeat at fixed intervals (e.g., increased ice cream sales every summer).
3. Noise/Residual: Random variations that cannot be explained by the trend or seasonality.



II. CLASSICAL MODELS VS. DEEP LEARNING

2.1. ARIMA (The Mathematical Foundation)

ARIMA (AutoRegressive Integrated Moving Average) is the "Gold Standard" for linear time series.

- AR (AutoRegression): Uses the relationship between an observation and a number of lagged observations.
- I (Integrated): Uses differencing of raw observations to make the time series stationary (mean and variance don't change over time).
- MA (Moving Average): Uses the dependency between an observation and a residual error from a moving average model.

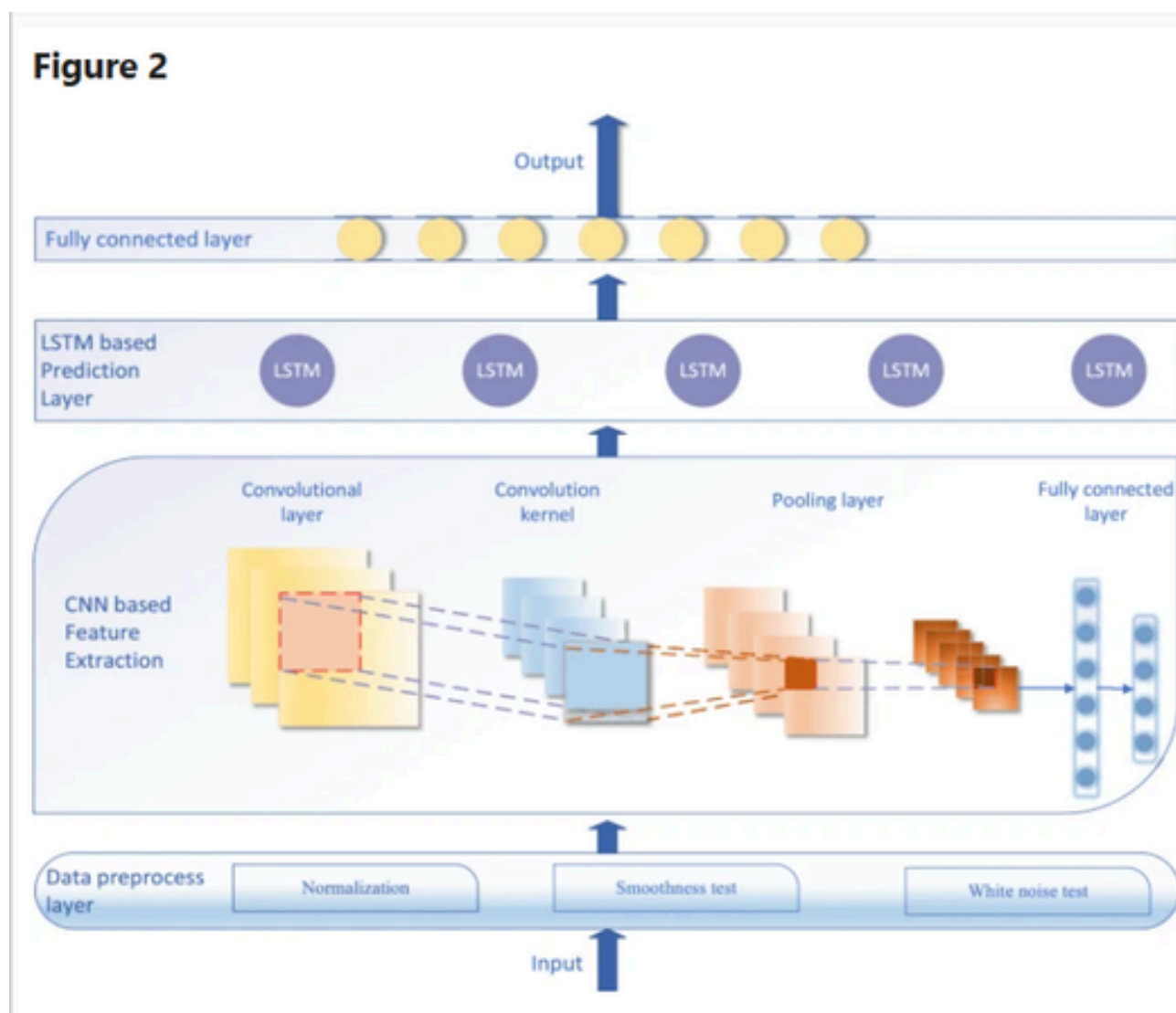
2.2. Prophet (The Industry Favorite)

Developed by Meta (Facebook), Prophet is designed for business forecasting. It handles missing data and outliers exceptionally well.

- Best for: Data with strong seasonal effects (weekly, yearly) and "holiday effects."
- Philosophy: It treats forecasting as a curve-fitting exercise rather than a time-dependency problem.

III. DEEP LEARNING FOR TIME SERIES: LSTM

When data is non-linear and highly complex, we use LSTMs (Long Short-Term Memory) networks.



3.1. Why LSTMs?

Standard Neural Networks have no "memory." Recurrent Neural Networks (RNNs) have short-term memory but suffer from the Vanishing Gradient problem (they forget long-term patterns). LSTMs use special "Gates" to decide what information to keep and what to throw away:

1. Forget Gate: Drops irrelevant info from the past.
2. Input Gate: Updates the current state with new info.
3. Output Gate: Decides what the next hidden state should be.

IV. PRACTICAL LAB: FORECASTING WITH PROPHET

In your project, you can use Prophet to predict how much storage your 50-file library will need in 6 months based on current upload rates.

Python

```
from prophet import Prophet
```

```
import pandas as pd
```

```
# 1. Prepare data (Columns must be 'ds' for date and 'y' for value)
```

```
df = pd.read_csv('system_load.csv') # ds: 2026-01-01, y: 0.75
```

```
# 2. Initialize and Fit the model
```

```
model = Prophet(yearly_seasonality=True, daily_seasonality=True)
```

```
model.fit(df)
```

```
# 3. Create a future dataframe for the next 30 days
```

```
future = model.make_future_dataframe(periods=30)
```

```
forecast = model.predict(future)
```

```
# 4. Plot the results
```

```
model.plot(forecast)
```

V. EVALUATION METRICS: HOW ACCURATE IS THE FUTURE?

You cannot use standard "Accuracy" for time series. Instead, we use error metrics:

- MAE (Mean Absolute Error): The average of the absolute differences between predicted and actual values.
- RMSE (Root Mean Square Error): Similar to MAE but punishes large errors more severely.
- MAPE (Mean Absolute Percentage Error): Expresses the error as a percentage (e.g., "Our prediction is 95% accurate").